

02/04

Pixel - smallest unit of a digital image or graphic that can be displayed on a screen.

Raster scan display - a common type of display system that generates images by scanning an electron beam across the screen, row by row, from top to bottom.

Frame buffer - Each pixel's color and intensity values are stored in the frame buffer, allowing the display to be refreshed by reading this data sequentially.

Resolution - no. of pixels.

line drawing algms

DDA - digital differential algorithm

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = m(x - (x+1))$$

$$y - y_1 = m(x - x - 1) = -m$$

$$y_1 = y + m$$

if $\Delta x > \Delta y$

$$x = x + 1$$

$$y = y + m$$

if $\Delta y > \Delta x$

$$y = y + 1$$

$$x = x + 1/m$$

1. form eqn of line (3,8) (8,11)

2. using DDA, find pts for the above line

$$y - y_1 = m(x - x_1)$$

$$8 - 11 = m(3 - 8)$$

$$m = 3/5 = 0.6$$

$$y - 8 = \frac{3}{5}(x - 3)$$

$$5y - 40 = 3x - 9$$

$$5y - 3x = 31$$

x	y	final pt
3	8	(3, 8)
4	8.6	(4, 9)
5	9.2	(5, 9)
6	9.8	(6, 10)
7	10.4	(7, 10)
8	11	(8, 11)

3. a) form eqn (3, 8) (6, 13)

b) DDA

for $y - y_1 = m(x - x_1)$ $\Delta x = 3, \Delta y = 5$ $\therefore \Delta y > \Delta x$, use eqn for incrementing y

$$y - 8 = \frac{5}{3}(x - 3) \Rightarrow y - 8 = \frac{5}{3}(x - 3)$$

$$3y - 24 = 5x - 15$$

$$3y - 5x = 9$$

for finding pts, take $y_1 = y + 1$

$$y - y_1 = m(x - x_1)$$

$$y - y - 1 = m(x - x_1)$$

$$m x - m x_1 = -1$$

$$x - x_1 = -1/m$$

$$x_1 = x + 1/m = x + (\frac{3}{5})^{-1} = x + 3/5$$

x y final pts

3 8 (3, 8)

3.6 9 (4, 9)

4.2 10 (4, 10)

4.8 11 (5, 11)

5.4 12 (5, 12)

6 13 (6, 13)

01/03/25

OpenGL programs

1. Quadrilateral printing

import math

import time

from OpenGL.GL import *

from OpenGL.GLU import *

from OpenGL.GLUT import *

x = -200

y = 0

def init():

glClearColor(0, 0, 0, 1)

gluOrtho2D(-300, 300, -300, 300)

def draw():

global x, y

glClearColor(GL_COLOR_BUFFER_BIT)

glColor3f(1, 0, 1)

glLineWidth(2)

glBegin(GL_QUADS)

glVertex2f(x, y)

glVertex2f(x+50, y)

glVertex2f(x+50, y+50)

glVertex2f(x, y+50)

glEnd()

glutSwapBuffers()

def main():

glutInit(sys.argv)

glutInitDisplayMode(GLUT_RGBA)

glutInitWindowSize(500, 500)

glutInitWindowPosition(600, 0)

glutCreateWindow("SAMPLE")

glutDisplayFunc(lambda: draw())

init()

glutMainLoop()

main()

A(-300, 0)

B(-150, 0)

C(-150, 50)

D(-300, 50)

Circle drawing

input: radii

input: line

Adv/Disadv: Disadvantage of DDA - m is usually float values - increases the cost/time for the calculations.

Bresenham Line drawing

Adv: only integer calculations and suitable for hardware implementation

Bresenham Algorithm

$x = x_1; y = y_1;$

Plot(x, y);

$P = 2\Delta y - \Delta x$

while $x < x_2$

{ $x++$;

if $P < 0$

$P = P + 2\Delta y$

Else

{ $y++$;

$P = P + 2\Delta y - 2\Delta x$

}

plot(x, y);

}

Q Find pts b/w (20, 10) & (30, 18) for plotting a line

Ans: $\Delta y = 8, \Delta x = 10$

PK x y

20 10

$P = 2\Delta y - \Delta x = 2 \times 8 - 10 = 6$

6 21 11

$P = P + 2\Delta y - 2\Delta x = 6 + 2 \times 8 - 2 \times 10 = 2$

2 22 12

$P = P + 2\Delta y - 2\Delta x = 2 + 2 \times 8 - 2 \times 10 = -2$

-2 23 12

$P = P + 2\Delta y = -2 + 2 \times 8 = 14$

14 24 13

10 25 14

6 26 15

2 27 16

-2 28 16

14 29 17

10 30 18

Q. Find points for line b/w (3, 8) and (3, 11)

Ans: $\Delta x = 5$, $\Delta y = 3$

pk x y

3 8

$$P = 2\Delta y - \Delta x = 2 \times 3 - 5 = 1$$

1 4 9

$$P = P + 2\Delta y - 2\Delta x = 1 + 2 \times 3 - 2 \times 5 = -3$$

-3 5 9

$$P = P + 2\Delta y = -3 + 2 \times 3 = 3$$

3 6 10

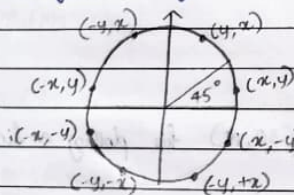
$$P = P + 2\Delta y - 2\Delta x = 3 + 2 \times 3 - 2 \times 5 = -1$$

-1 7 10

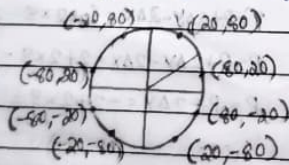
$$P = P + 2\Delta y = -1 + 2 \times 3 = 5$$

5 8 11

Circle generation (Symmetry)



Find the 7 pts that can be found from the point (20, 80)



16/03/25

Midpoint Circle Algorithm

$x = 0$; $y = 0$;

plot($x + x_c$, $y + y_c$);

$p = 5/4 - x$; ($0, 1 - 0$)

while $x < y$

{ $x = x + 1$;

if $P < 0$ $p = p + 2x + 1$

else { $y = y + 1$; $p = p + 2x - 2y + 1$;

plot($x + x_c$, $y + y_c$);

Determine Symmetry points in the other seven quadrants.

Add centre coordinates x_c, y_c to the coordinates found.

}

Q. Radius = 5 Centre (10, 20)

Ans:

P x' y' x y

$P < 0$ $P = P + 2x + 1$

0 5 10 25

else $P = P + 2x - 2y + 1$

-4 1 5 11 25

$P = 1 - 8 = 1 - 5 = -4$

-1 2 5 12 25

$P = P + 2x + 1 = -4 + 2 \times 1 + 1 = -1$

4 3 4 13 24

3 4 3 14 23

Q $\delta = 8$ center $(10, 20)$ first pt $= (10, 28)$

Ans:

pk	x'	y'	x	y
	0	8	10	28
-7	1	8	11	28
-4	2	8	12	28
1	3	7	13	27
-6	4	7	14	27
3	5	6	15	26
2	6	5	16	25

$$P = 1 - 8 = -7$$

$$P = P + 2x + 1 = -7 + 2 + 1 = -4$$

$$P = -4 + 4 + 1 = 1$$

$$P = P + 2x - 2y + 1 = 1 + 6 - 14 + 1 = -6$$

$$P = P + 2x + 1 = -6 + 8 + 1 = 3$$

$$P = P + 2x - 2y + 1 = 3 + 10 - 12 + 1 = 2$$

Shape Filling Algorithms

Flood fill:

Floodfill (int x, y , fillcolor, old color)

{ if getpixel(x, y) = oldcolor

{ setpixel(x, y , fillcolor)

Floodfill($x+1, y$, fill, old)

Floodfill($x-1, y$

Floodfill($x, y+1$, fill, old)

Floodfill($x, y-1$, fill, old)

}

}

Boundary fill:

boundaryfill (int x, y , fillcolor, bordercolor)

{ getpixel(x, y , interiorcolor)

if (interiorcolor != bordercolor) && (interiorcolor != fillcolor)

{ setpixel(x, y , fillcolor)

boundaryfill($x+1, y$, fillcolor, bordercolor)

boundaryfill($x-1, y$, fillcolor, bordercolor)

boundaryfill($x, y+1$, fillcolor, bordercolor)

boundaryfill($x, y-1$, fillcolor, bordercolor)

}

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Module 22D transformations:

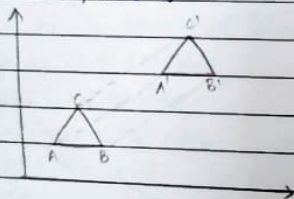
1. Translation
2. Scaling
3. Rotation

1. Helps to move an object from 1 location to another without affecting the size or shape.

Given x, y

t_x, t_y

$$x' = x + t_x, \quad y' = y + t_y$$



2. Helps to scale the object to a predefined scale (enlarge, compress etc.)

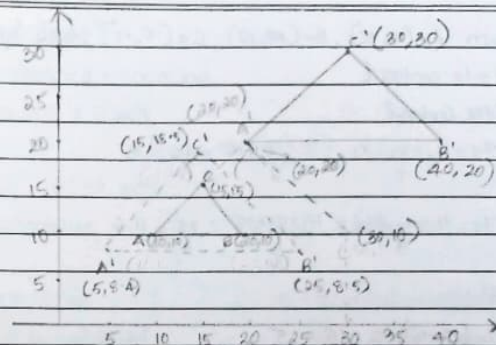
Given x, y

s_x, s_y

$$x' = x * s_x$$

$$y' = y * s_y$$

$$\text{where } s_x = s_y = 2$$



for scaling about a ref point

1. translation $(-t_x, -t_y)$

2. scaling about origin

3. Retranslation $(+t_x, +t_y)$

Given: x, y ; s_x, s_y

$$\text{then, } x' = (x - t_x) s_x + t_x$$

$$y' = (y - t_y) s_y + t_y$$

Q Given $A = (10, 10)$, $B = (20, 10)$, $C = (15, 15)$

$$s_x = s_y = 2, (x, y) = (10, 10) \rightarrow (\text{w.r.t } A)$$

Ans:

finding B'

$$x' = (20 - 10) * 2 + 10 = 30$$

$$y' = (10 - 10) * 2 + 10 = 10$$

$$B' = (30, 10)$$

finding C'

$$x' = (15 - 10) * 2 + 10 = 20$$

$$y' = (15 - 10) * 2 + 10 = 20$$

$$C' = (20, 20)$$

Q Given $A = (10, 10)$, $B = (20, 10)$, $C = (15, 15)$ scale by 2

is to centroid

Since centroid

$$x_1 = \frac{x_1 + x_2 + x_3}{3} = \frac{10 + 20 + 15}{3} = \frac{45}{3} = 15$$

$$y_1 = \frac{y_1 + y_2 + y_3}{3} = \frac{10 + 10 + 15}{3} = \frac{35}{3} = 11.6$$

A'

$$x' = (10 - 15) \times 2 + 15 = 5$$

$$y' = (10 - 11.6) \times 2 + 11.6 = 8.4$$

$$A' = (5, 8.4)$$

B' :

$$x' = (20 - 15) \times 2 + 15 = 25$$

$$y' = (10 - 11.6) \times 2 + 11.6 = 8.4$$

$$B' = (25, 8.4)$$

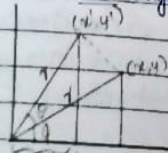
C' :

$$x' = (15 - 15) \times 2 + 15 = 15$$

$$y' = (15 - 11.6) \times 2 + 11.6 = 18.4$$

$$C' = (15, 18.4)$$

Rotation about origin $(x, y, 0)$



$$x' = r \cos(\theta + \phi)$$

$$= r(\cos\theta \cos\phi - \sin\theta \sin\phi)$$

$$= r \cos\theta \cos\phi - r \sin\theta \sin\phi$$

$$= r \cos\theta \cos\phi - r \sin\theta \sin\phi$$

$$y' = r \sin(\theta + \phi)$$

$$= r \sin\theta \cos\phi + r \cos\theta \sin\phi$$

$$= r \sin\theta \cos\phi + r \cos\theta \sin\phi$$

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Q $A(20, 10)$ $B(50, 35)$

i) 30° anticlockwise

A' :

$$x' = x \cos\theta - y \sin\theta$$

$$= 20 \cos 30^\circ - 10 \sin 30^\circ$$

$$= 12.32$$

$$y' = x \sin\theta + y \cos\theta$$

$$= 20 \sin 30^\circ + 10 \cos 30^\circ$$

$$= 18.66$$

$$A' (12.32, 18.66)$$

B' :

$$x' = x \cos\theta - y \sin\theta$$

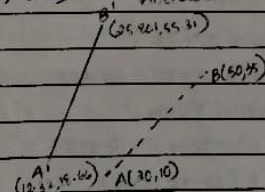
$$= 50 \cos 30^\circ - 35 \sin 30^\circ = 25.801$$

$$y' = x \sin\theta + y \cos\theta$$

$$= 50 \sin 30^\circ + 35 \cos 30^\circ = 55.31$$

$$B' (25.801, 55.31)$$

Anticlockwise



ii) 30° clockwise

A'

$$x' = x \cos(-\theta) - y \sin(-\theta) = x \cos\theta + y \sin\theta$$

$$= 20 \cos 30^\circ + 10 \sin 30^\circ = 22.32$$

$$y' = x \sin(-\theta) + y \cos(-\theta) = -x \sin\theta + y \cos\theta$$

$$= -20 \sin 30^\circ + 10 \cos 30^\circ = -1.339$$

$$A' (22.32, -1.339)$$

B' :

$$x' = x \cos(-\theta) - y \sin(-\theta) = x \cos\theta + y \sin\theta$$

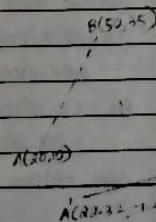
$$= 50 \cos 30^\circ + 35 \sin 30^\circ = 60.301$$

$$y' = x \sin(-\theta) + y \cos(-\theta) = -x \sin\theta + y \cos\theta$$

$$= -50 \sin 30^\circ + 35 \cos 30^\circ = 5.310$$

$$B' (60.301, 5.310)$$

Clockwise



Rotation about a particular point (x_1, y_1)

this involves $\rightarrow$ - translation

- rotation

- re-translation

$$x' = (x - x_1) \cos \theta - (y - y_1) \sin \theta + x_1$$

$$y' = (x - x_1) \sin \theta + (y - y_1) \cos \theta + y_1$$

Q. Rotate $A(20, 10)$ $B(50, 35)$ w.r.t. A

i) anticlockwise 30°

$$x_1 = 20, y_1 = 10$$

$$x'_A = 20, y'_A = 10$$

$$x'_B = (x - x_1) \cos \theta - (y - y_1) \sin \theta + x_1$$

$$= (50 - 20) \cos 30 - (35 - 10) \sin 30 + 20$$

$$= 30 \times \frac{\sqrt{3}}{2} - 25 \times \frac{1}{2} + 20 = 33.480$$

$$y'_B = (x - x_1) \sin \theta + (y - y_1) \cos \theta + y_1$$

$$= (50 - 20) \sin 30 + (35 - 10) \cos 30 + 10$$

$$= 46.650$$

after rotation $A'(20, 10)$, $B'(33.48, 46.65)$

ii) clockwise 30°

$$x_1 = 20, y_1 = 10$$

$$x'_B = (x - x_1) \cos(-\theta) - (y - y_1) \sin(-\theta) + x_1$$

$$= (50 - 20) \cos 30 + (35 - 10) \sin 30 + 20 = 58.480$$

$$y'_B = (x - x_1) \sin(-\theta) + (y - y_1) \cos(-\theta) + y_1$$

$$= -(50 - 20) \sin 30 + (35 - 10) \cos 30 + 10 = 16.650$$

after rotation $A'(20, 10)$, $B'(58.48, 16.65)$

Matrix Product

$\begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} & \\ & \end{bmatrix}$

translation coordinate

translation $\rightarrow x' = x + tx$; $y' = y + ty$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + tx \\ y + ty \\ 1 \end{bmatrix}$$

$3 \times 3 \quad 3 \times 1$

$$\begin{bmatrix} x' & y' & 1 \end{bmatrix} = \begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ tx & ty & 1 \end{bmatrix} = \begin{bmatrix} x + tx & y + ty & 1 \end{bmatrix}$$

$1 \times 3 \quad 3 \times 3$

scaling about origin

$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} S_x \cdot x \\ S_y \cdot y \end{bmatrix}$$

homogeneous form 3×3

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x \cdot x \\ S_y \cdot y \\ 1 \end{bmatrix}$$

* homogeneous form allows

Composite transformation $\rightarrow$ even

100 transⁿ pnt can be found in 1

step by forming single matrix - less work

scaling about (x_1, y_1)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_1 \\ 0 & 1 & y_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_1 \\ 0 & 1 & -y_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

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Rotation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

for rotation about (x_1, y_1)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_1 \\ 0 & 1 & y_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_1 \\ 0 & 1 & -y_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

a) write the matrix form for: (Homogeneous)

1) Translation $T_x=5$ $T_y=5$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x+5 \\ y+5 \\ 1 \end{bmatrix}$$

2) Rotation about $A(10,10)$ i) 30° anticlockwise

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

ii) 30° clockwise

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-30) & -\sin(-30) & 0 \\ \sin(-30) & \cos(-30) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3. scale $S_x=2$ $S_y=3$

i) scale about origin

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 2x \\ 3y \\ 1 \end{bmatrix}$$

ii) scale about $(5,5)$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Reflection

* About x axis

$$x' = x, y' = -y$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* About y axis $\rightarrow x' = -x, y' = y$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* About $y = x \rightarrow x' = y, y' = x$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* About $y = -x$

$$x' = -y, y' = -x$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

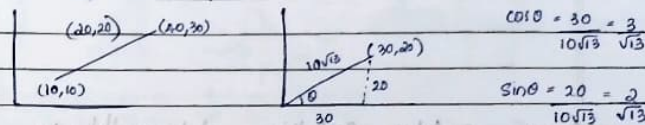
* Reflection about origin

$$x' = -x, y' = -y$$

Q. write the series of matrices for the reflection of $(20, 20)$ about a line with end points $(10, 10)$ $(40, 30)$. After translation points $(0, 0)$ $(30, 20)$

$$\cos \theta = 3/\sqrt{13}, \sin \theta = 2/\sqrt{13}$$

Ans:



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

20/1/25

$$\begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 \\ \sin \theta_2 & \cos \theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

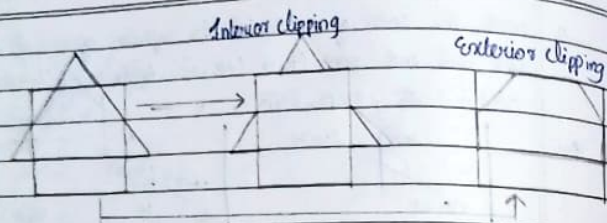
$$= \begin{bmatrix} \cos \theta_2 \cos \theta_1 - \sin \theta_2 \sin \theta_1 & -\cos \theta_2 \sin \theta_1 - \sin \theta_2 \cos \theta_1 & 0 \\ \sin \theta_2 \cos \theta_1 + \cos \theta_2 \sin \theta_1 & -\sin \theta_2 \sin \theta_1 + \cos \theta_2 \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Clipping

Interior clipping - removes portions inside a window

Exterior clipping - removes portions outside a window



Types

1. Point clipping $x_L \leq x \leq x_R, y_B \leq y \leq y_T$
2. Line clipping
3. Polygon
4. Curve

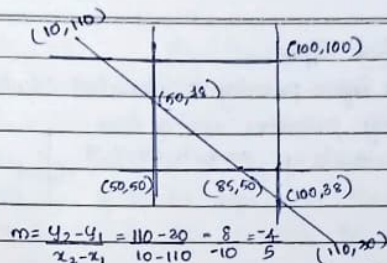
Line clipping (Cohen Sutherland algorithm)

clip a line against a rectangle

1001	1000	1010	bit 1 is set if $x < x_L$
0001	0000	0010	bit 2 is set if $x > x_R$
0101	0100	0110	bit 3 is set if $y < y_B$
			bit 4 is set if $y > y_T$

$x \leq x_L$	$x \geq x_R$	1010	$x_L - x_{left}$
$y \leq y_B$	(x_L, y_B)	$y_B - y_{bottom}$	$y_B - y_{top}$
$y \geq y_T$	(x_R, y_T)		
(x_L, y_B)			$y = mx + c$

a)



steps:

find end point codes

check for inside, outside

find required intersection

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{110 - 20}{10 - 110} = \frac{90}{-100} = -0.9$$

$$\text{line eqn. } (y - y_1) = m(x - x_1)$$

$$y = -0.9x + 110$$

$$y - 110 = -0.9(x - 10)$$

$$= 28 \text{ when } x = 100$$

$$y = -0.9x + 110$$

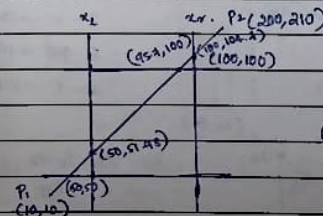
$$50 = -0.9x + 110 \text{ when } y = 50$$

$$y = -0.9x + 110$$

$$x = \frac{50 - 110}{-0.9} = \frac{-60}{-0.9} = 66.67$$

$$y = 28 \text{ when } x = 100$$

a)



$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{210 - 10}{190 - 19} = \frac{200}{171} = 1.17$$

$$(y - y_1) = m(x - x_1)$$

$$y - 10 = 1.17(x - 10)$$

$$y - 10 = 1.17x - 11.7$$

$$y = 1.17x - 1.7$$

$$y = 1.05x - 0.52$$

when $y = 100$

$$100 = 1.05x - 0.52$$

when $x = 50$

when $y = 50$

$$x = \frac{100 + 0.52}{1.05} = 95.7$$

$$y = 1.05x - 0.52$$

$$x = \frac{50 + 0.52}{1.05} = 48.2$$

$$= 51.48$$

$$1.05$$

Long Bounding Line Clipping

more line testing done before proceeding to intersection calculations.

Fastest line clipping using parametric eq of lines

$$x = x_1 + u \Delta x \quad 0 \leq u \leq 1, \quad \Delta x = x_2 - x_1, \Delta y = y_2 - y_1$$

$$y = y_1 + u \Delta y$$



parametric line eqn

$$(20, 10) \quad (60, 80)$$

when $u = 0.8$

when $u = 0.2$

$$\Delta x = 60 - 20 = 40$$

$$x = 20 + 0.8 \times 40 = 52$$

$$x = 20 + 0.2 \times 40 = 28$$

$$\Delta y = 80 - 10 = 70$$

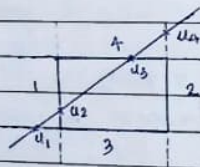
$$y = 10 + 0.8 \times 70 = 66$$

$$y = 10 + 0.2 \times 70 = 24$$

$$x = 20 + u \Delta x$$

$$y = 10 + u \Delta y$$

06/08/25



1 - out $\rightarrow$ in

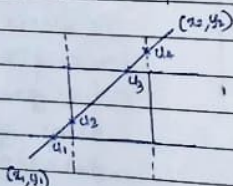
2 - in $\rightarrow$ out

3 - out $\rightarrow$ in

4 - in $\rightarrow$ out

$$u_1 = \frac{y_2 - y_1}{\Delta y}$$

$$u_2 = \frac{x_2 - x_1}{\Delta x}$$



$$u_3 = \frac{y_2 - y_1}{\Delta y}$$

$$u_4 = \frac{x_2 - x_1}{\Delta x}$$

$$UP_K < q_K \quad K = 1, 2, 3, 4$$

$$P_i = -\Delta x \quad q_i = x_1 - x_{\min}$$

$$P_2 = \Delta x$$

$$q_2 = x_{\max} - x_1$$

$$P_3 = -\Delta y$$

$$q_3 = y_1 - y_{\min}$$

$$P_4 = \Delta y$$

$$q_4 = y_{\max} - y_1$$

Any line parallel to the clipping boundary will have $P_i = 0$ where i is the edge. If for the edge $q_i < 0$ then the line is outside the edge. If $q_i \geq 0$ the line is inside the boundaries.

Q. window boundaries (10, 10) (50, 50)

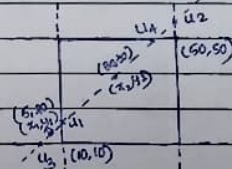
Line (5, 20) (30, 30)

$$P_1 = -25$$

$$P_2 = 25$$

$$P_3 = -10$$

$$P_4 = 10$$



$$q_1 = x_1 - x_{\min} = -5$$

$$q_2 = x_{\max} - x_1 = 50 - 5 = 45$$

$$q_3 = y_1 - y_{\min} = 20 - 10 = 10$$

$$q_4 = y_{\max} - y_1 = 50 - 20 = 30$$

$$u_1 = -5/-25 = 1/5$$

$$u_2 = q_1/P_1$$

$$u_2 = 45/25 = 9/5$$

$$u_3 = -10/-10 = 1$$

$$u_4 = 30/10 = 3$$

at point $u = 0.2$
 $x = 10$

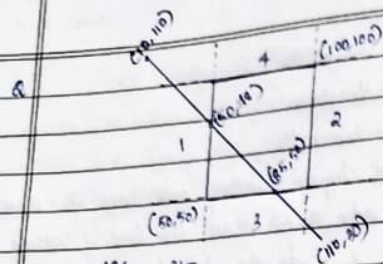
$$y = 10 + u \Delta y$$

$$\text{out} \rightarrow \text{in} \rightarrow \max(y_5, -1, 0) = 15 \times 0.2$$

$$= 20 + 0.2 \times 10$$

$$\text{in} \rightarrow \text{out} \rightarrow \min(9/5, 3, 1) = 1$$

$$= 20 + 2 = 22$$



$$\begin{aligned} P_1 &= -100 & Q_1 &= 10 - 50 = -40 \\ P_2 &= 100 & Q_2 &= 100 - 10 = 90 \\ P_3 &= 60 & Q_3 &= 110 - 50 = 60 \\ P_4 &= -80 & Q_4 &= 100 - 110 = -10 \end{aligned}$$

$$C_1 = -40 / -100 = 2/5$$

$$u_2 = 90 / 100 = 0.9 \quad \text{cut} \rightarrow \text{in max}(2/5, 1/5, 0) = 2/5$$

$$u_3 = 60 / 60 = 1 \quad \text{in} \rightarrow \text{out min}(0.9, 3/4, 1) = 3/4$$

$$u_4 = -10 / -80 = 1/8$$

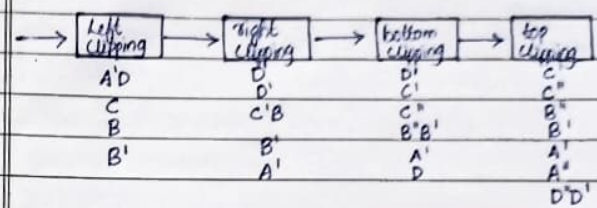
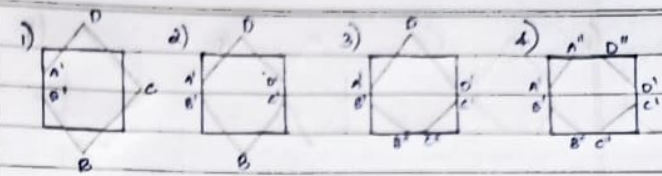
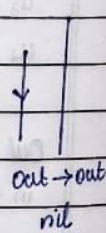
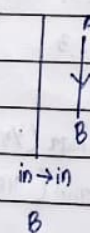
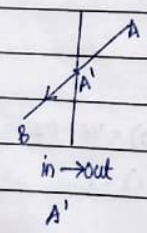
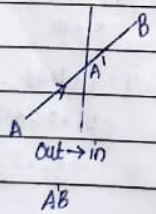
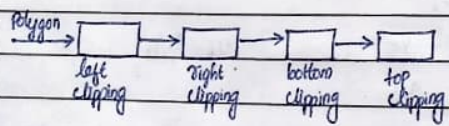
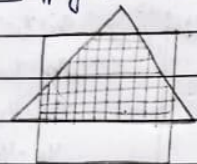
$$\text{At } u_1 = 2/5$$

$$\text{At } u_2 = 3/4$$

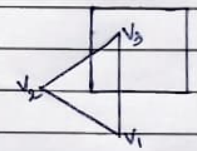
$$y = 110 + 2 \times -80 = 110 - 160 = -50$$

$$x = 10 + 3 \times 100 = 10 + 300 = 310$$

Polygon Clipping (Sutherland-Hodgman Algorithm)



Q. clip the fig w.r.t rectangular window



Ans:

